JSS COLLEGE OF ARTS, COMMERCE AND SCIENCE

An Autonomous College, Affiliated to University of Mysore Re-accredited by NACC with 'A+' Grade CGPA: 3.30 Ooty Road, Mysuru – 570025

A

Project Report On

"IOT BASED WATER LEVEL MONITORING SYSTEM"

Submitted in the partial fulfilment of the requirements for the award of

IV Semester

Masters of Science in Computer Science

Submitted by

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Under the guidance of

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570025 2025-2026

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CERTIFICATE

This is to certify that, **Tejaswini. K (P01BE24S038048)**, has successfully completed the Project work entitled **"IoT Based Water Level Monitoring System"** and submitted the report in partial fulfilment of the requirement of **IV Semester, Master of Science in Computer Science by JSS College of Arts, Commerce and Science, Mysuru**, during the academic year **2025-2026**. It is to certified that all corrections and suggestions, including for the internal assessment, have been incorporated in the report. This report has been approved as it satisfies the academic requirements in respect of Project work prescribed for IV Semester M.Sc.

Signature of the Guide **Signature of the HOD**

Date of Examination:

VALUED

Name and Signature of the Examiners:

1._____

2._____

DECLARATION

I, **Tejaswini. K (P01BE24S038048)**, hereby declare that the Project work entitled **"IoT Based Water Level Monitoring System"** submitted to the **JSS College of Arts, Commerce and Science, Ooty Road, Mysuru (Affiliated to University of Mysore)** during the academic year **2025-2026**, is a record of an original work done by me under the guidance of **Mrs. SUMANASHREE Y S** H.O.D and Asst. Professor, PG Department of Computer Science.

Tejaswini. K [P01BE24S038048] **Signature**

Date:

Place:

ACKNOWLEDGEMENT

It gives me immense pleasure to present before you my Project work titled **"IoT Based Water Level Monitoring System"**. The joy and satisfaction that accompany the successful completion of any task would be incomplete without the mention of those who made it possible. I am glad to express my gratitude towards our prestigious institution **JSS COLLEGE OF ARTS, COMMERCE & SCIENCE, OOTY ROAD, MYSORE**, providing us with utmost knowledge, encouragement and the maximum facilities in undertaking this project.

I wish to express our sincere thanks to our respected principal **Dr. PRABHU M** for all their support.

I wish to express our deepest gratitude and special thanks to **Mrs. SUMANASHREE Y S**, Prof & H.O.D, PG Department of Computer Science, for all her guidance and encouragement.

We sincerely acknowledge the guidance and constant encouragement of our Project work guide, H.O.D and Asst. Prof. **Mrs. POOJA**

Tejaswini. K (P01BE24S038048)

CHAPTER 1

ABSTRACT

The Internet of Things (IOT) has enabled the development of smart and automated systems for efficient resource management. This project presents an IOT-Based Water Level Monitoring and Motor Control System that provides real-time monitoring of water levels along with both manual and automatic control of a water pump using a web-based interface.

The system uses a microcontroller such as NodeMCU ESP8266 and an ultrasonic sensor to measure the water level in a tank. The measured data is processed and displayed on a web page, allowing users to monitor the water level remotely from any device connected to the internet. Unlike systems that rely on mobile applications like Blynk, this project uses a dedicated web server for data visualization and control.

A key feature of this system is its dual-mode operation: manual and automatic. In manual mode, the user can turn the motor ON or OFF using buttons available on the web page. In automatic mode, the system intelligently controls the motor based on predefined water level thresholds. When the water level drops below a minimum limit, the motor automatically turns ON, and when the tank becomes full, the motor turns OFF, preventing overflow and water wastage.

This system improves efficiency, reduces human effort, and ensures better water management. It is cost-effective, easy to implement, and suitable for domestic as well as industrial applications. The project demonstrates how IoT and web technologies can be integrated to create a smart and user-friendly solution for real-time monitoring and automation.

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CHAPTER 1

INTRODUCTION

Water is an essential resource required for daily life, agriculture, and industrial processes. Efficient management of water resources is becoming increasingly important due to rising demand and limited availability. Conventional water tank systems rely on manual monitoring and motor control, which often leads to problems such as water overflow, wastage, and dry running of pumps.

To overcome these limitations, modern technologies like the Internet of Things (IoT) provide smarter and more efficient solutions. IoT enables real-time monitoring and control of devices through internet connectivity. This project focuses on developing an IoT-Based Water Level Monitoring and Motor Control System that not only displays the water level but also allows both manual and automatic control of the motor using a web interface.

The system is built using a microcontroller such as NodeMCU ESP8266, which has built-in Wi-Fi capability. An ultrasonic sensor is used to measure the water level in the tank. The collected data is processed and hosted on a web server, where it is displayed in a user-friendly format. Users can access this web page through a browser to check the water level and control the motor.

One of the important features of this system is the inclusion of both manual and automatic modes. In manual mode, the user can operate the motor using ON and OFF buttons on the web page. In automatic mode, the system controls the motor based on predefined water levels. When the water level falls below a certain threshold, the motor is automatically turned ON, and when the tank reaches its maximum level, the motor is turned OFF. This prevents water overflow and ensures efficient usage.

Unlike other IoT systems that depend on mobile applications such as Blynk, this project uses a web-based approach, making it more flexible and accessible from any device without installing additional applications.

In conclusion, this system provides a smart, reliable, and efficient solution for water level monitoring and motor control. It reduces manual effort, improves water conservation, and demonstrates the practical implementation of IoT in everyday life.

1.1 OBJECTIVES

The main objectives of the IoT-Based Water Level Monitoring System are:

1. To develop an automated system for monitoring water levels in tanks and reservoirs.
2. To measure water levels accurately using ultrasonic sensors and microcontroller technology.
3. To provide real-time monitoring of water levels through IoT platforms such as mobile applications or web dashboards.
4. To reduce water wastage caused by tank overflow and inefficient manual monitoring.
5. To send notifications or alerts when the water level reaches minimum or maximum threshold values.
6. To improve efficient water management in homes, industries, and agricultural fields.
7. To minimize human effort by enabling remote monitoring through internet connectivity.
8. To create a low-cost, reliable, and energy-efficient smart water monitoring system.

1.2 PROBLEM STATEMENT

Water is one of the most important natural resources, but improper monitoring and management of water tanks often lead to overflow, water wastage, and shortage issues. In many homes, industries, and agricultural systems, water levels are checked manually, which is time-consuming and inefficient. Manual monitoring may also result in delayed actions, causing unnecessary water loss and damage to pumping systems.

Existing traditional systems lack real-time monitoring, remote accessibility, and automated alert mechanisms. Therefore, there is a need for a smart and efficient solution that can continuously monitor water levels and provide instant updates to users.

The proposed IoT-Based Water Level Monitoring System addresses these problems by using sensors, microcontrollers, and internet connectivity to monitor water levels in real time and notify users whenever the tank becomes full or empty.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

IoT-based water level monitoring systems have become an important solution for efficient water management in homes, agriculture, and industries. These systems help in real-time monitoring of water levels and automation of pump control, reducing human effort and preventing water wastage. This literature survey explores various existing systems and research works related to water level monitoring, IoT-based automation, sensor technologies, and web-based control systems.

2.2 IoT Based Water Level Monitoring System Using NodeMCU

AUTHOR'S: R. S. Patil, A. Kumar
YEAR: 2021

FINDINGS:

This project focuses on using NodeMCU ESP8266 and ultrasonic sensors for real-time water level monitoring. The system sends data to a cloud platform and enables remote monitoring through the internet. It also includes automatic motor control to prevent overflow and dry running of pumps. The study shows that IoT-based systems improve efficiency and reduce manual intervention in water management.

2.3 Smart Water Level Monitoring and Control System

AUTHOR'S: S. K. Sharma, P. Reddy
YEAR: 2022

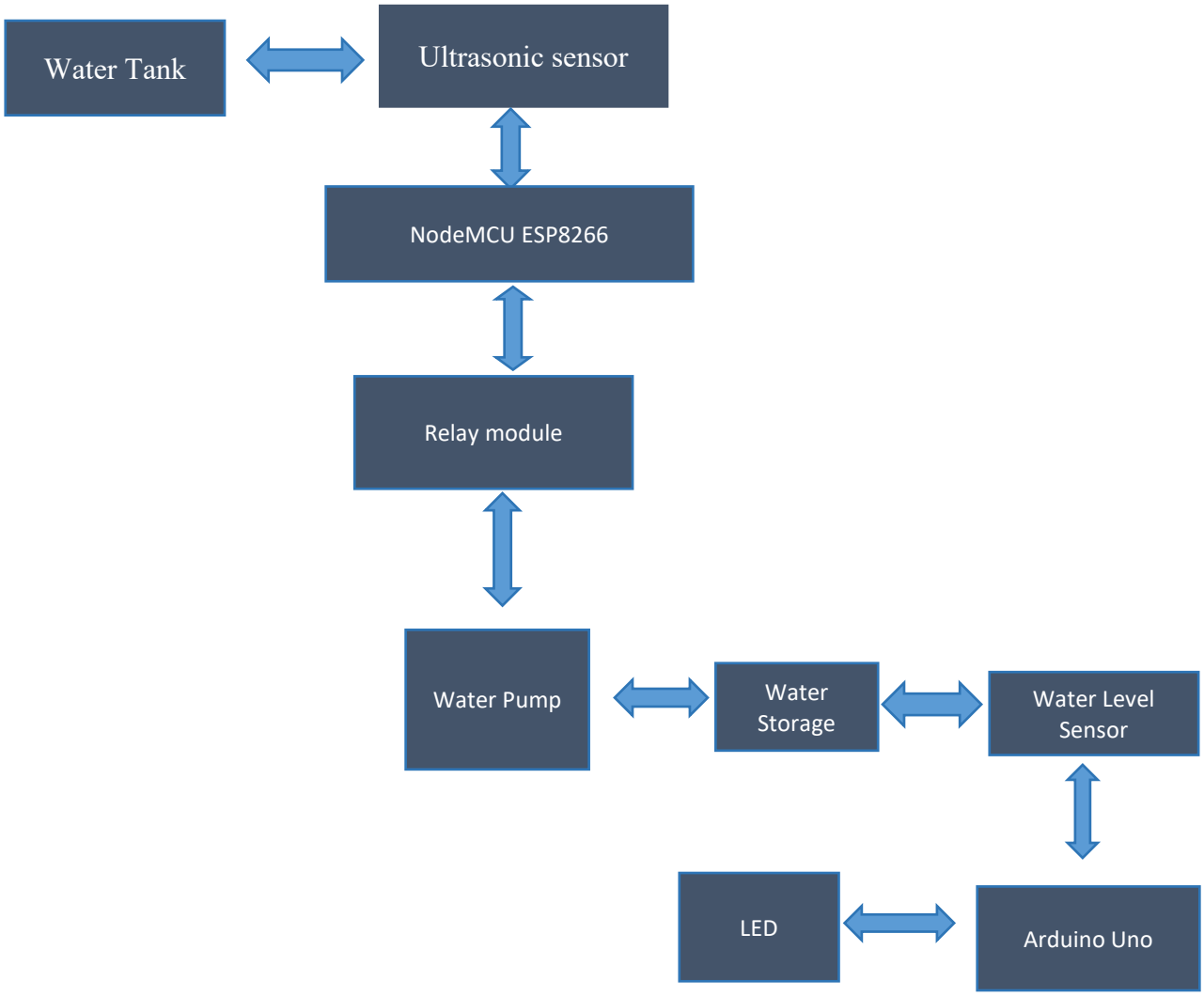
FINDINGS:

This research proposes a smart water tank monitoring system using an HC-SR04 Ultrasonic Sensor and microcontroller. The system provides automatic ON/OFF control of water pumps based on predefined water levels. It helps in preventing water wastage and ensures proper utilization of water resources. The system also provides real-time updates and alerts to users.

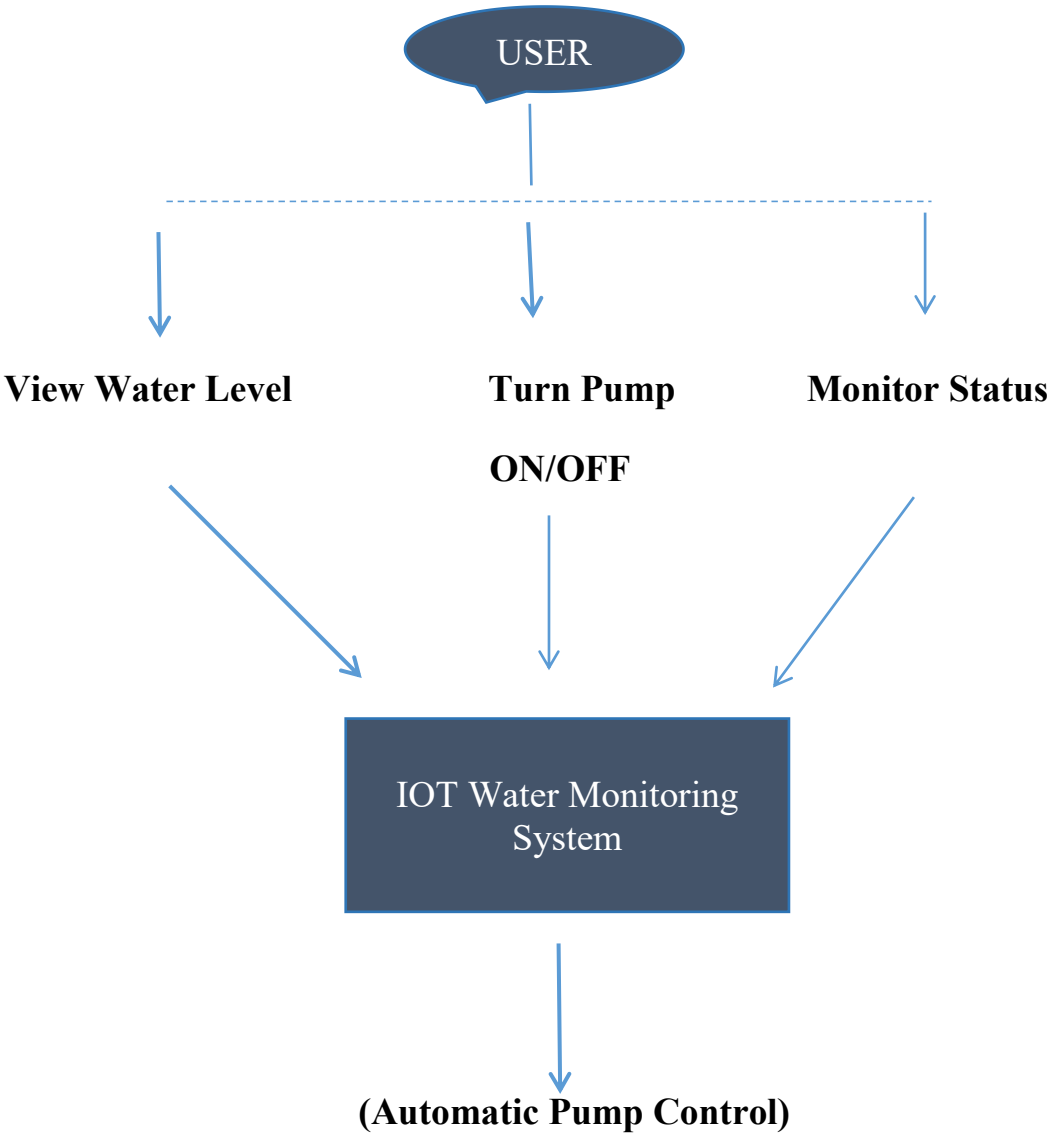
CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE



3.2 USECASE DIAGRAM



3.3 COMPONENTS

1. NodeMCU ESP8266

Description:

- Main microcontroller with built-in Wi-Fi
- Handles sensor data, web server, and motor control



2. HC-SR04 Ultrasonic Sensor

Description:

- Measures distance to water surface
- Used to calculate water level



3. Relay Module 5V

Description:

- Acts as a switch to control the motor
- Controlled by NodeMCU



4. Submersible Water Pump

Description:

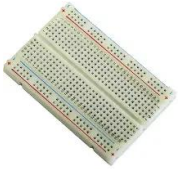
- Pumps water into the tank
- Turns ON/OFF based on system logic



5. Breadboard

Description:

- Used for circuit connections without soldering



6. Jumper Wires

Description:

- Connect components together



7. Power Supply Module

Description:

- Provides required voltage to NodeMCU and components



CHAPTER 4

REQUIREMENTS

4.1 HARDWARE COMPONENTS

Component	Purpose
NodeMCU/ESP8266	Microcontroller with Wi-Fi
Ultrasonic Sensor (HC-SR04)	Measure water level
Jumper Wires	Connections
Breadboard	Circuit setup
Power Supply	System power

4.2 SOFTWARE COMPONENTS

Software	Purpose
Arduino IDE	Programming
Blynk / ThingSpeak	IoT cloud platform
Embedded C/C++	Coding language

CHAPTER 5

WORKING PRINCIPLE

Introduction

The IoT Water Level Monitoring and Control System is a smart water management system used to monitor the water level in a storage tank and control the water pump automatically or manually through a web page. The system uses sensors, a microcontroller, internet connectivity, and a relay module to provide real-time monitoring and remote control functionality.

The main objective of this project is to reduce water wastage, avoid overflow of tanks, minimize manual effort, and provide efficient water management using Internet of Things technology.

Working Principle

The working principle of the proposed system is divided into different stages:

1. Water Level Detection

The system uses an ultrasonic sensor or water level sensor to measure the amount of water present in the tank. The sensor is placed at the top of the tank and continuously detects the distance between the sensor and the water surface.

The ultrasonic sensor sends ultrasonic sound waves toward the water surface and receives the reflected echo signal.

The distance is calculated using:

Distance=Speed of Sound×Time Taken2\text{Distance} = \frac{\text{Speed of Sound} \times \text{Time Taken}}{2}Distance=2Speed of Sound×Time Taken

After calculating the distance, the water level is determined by comparing the measured distance with the height of the tank.

Example:

- Small distance → High water level
- Large distance → Low water level

The measured water level data is sent to the microcontroller for further processing.

2. Microcontroller Processing

The heart of the system is the microcontroller such as:

- Arduino UNO
- NodeMCU
- ESP8266/ESP32 from Espressif Systems

The microcontroller performs the following functions:

- Reads sensor values
- Calculates water level percentage
- Sends data to the web server
- Controls relay operation
- Handles automatic and manual control

The controller continuously checks whether the water level is LOW, MEDIUM, or FULL.

3. Internet Connectivity and IoT Operation

The microcontroller is connected to the internet through Wi-Fi. The water level information is uploaded to a web server or cloud platform.

Using IoT technology, the user can monitor the tank condition remotely through:

- Mobile phone
- Laptop
- Computer
- Web browser

The data is updated in real time on the web page.

4. Web Page Monitoring System

A web page is developed to display the current water level status and provide control options for the user.

The web page displays:

- Water level percentage
- Tank condition
- Motor status
- ON/OFF buttons

Water Level Indication

The water level may be displayed in:

- Percentage form
- Graphical bar representation
- Text indication

Example:

- 0–25% → Low
- 26–75% → Medium
- 76–100% → Full

The web page allows users to monitor the system from any location with internet access.

5. Manual Motor Control

The system includes ON and OFF buttons on the web page for manual control of the water pump.

Motor ON Operation

When the user presses the ON button:

1. Signal is sent from the web page to the microcontroller.
2. The microcontroller activates the relay module.
3. The relay switches ON the water pump.

Motor OFF Operation

When the OFF button is pressed:

1. Signal is sent to the microcontroller.
2. Relay is deactivated.
3. Water pump turns OFF.

This feature allows remote control of the motor through the internet.

6. Automatic Pump Control

The system automatically controls the motor depending on the water level in the tank.

Automatic ON Condition

If the water level falls below the minimum threshold:

- The controller automatically switches ON the pump.

Automatic OFF Condition

If the tank becomes full:

- The controller automatically switches OFF the pump.

This automatic operation helps in:

- Preventing tank overflow
- Reducing water wastage
- Protecting the motor from dry running
- Saving electricity
- Eliminating continuous human monitoring

7. Relay Module Operation

A relay module is used as an electronic switch between the microcontroller and the water pump motor.

The microcontroller operates at low voltage, while the motor requires high voltage. Therefore, the relay safely controls the motor using low-power control signals.

Relay Functions:

- Isolates low-voltage and high-voltage circuits
- Provides safe motor switching
- Controls AC water pump operation

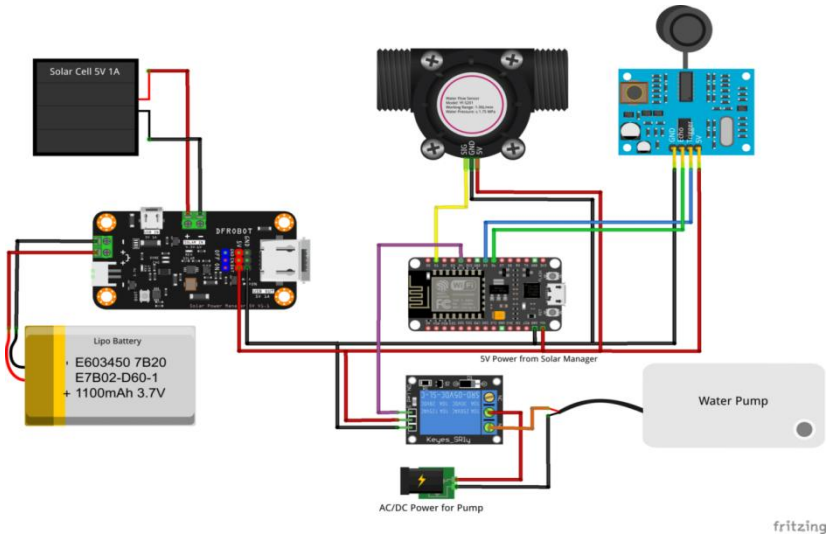
8. System Flow Operation

The overall operation of the system can be explained as follows:

1. Sensor measures water level.
2. Microcontroller receives sensor data.
3. Water level is calculated.
4. Data is uploaded to the web page.
5. User monitors tank status online.
6. Pump can be controlled:
    - Automatically based on level
    - Manually through web buttons
7. Relay switches motor ON/OFF.

CHAPTER 6

CONNECTION DIAGRAM



The IoT-Based Water Level Monitoring and Motor Control System is built using a NodeMCU ESP8266 microcontroller, an HC-SR04 Ultrasonic Sensor, and a Relay Module 5V to control the water pump.

The ultrasonic sensor is used to measure the water level in the tank. It has four pins: VCC, GND, TRIG, and ECHO. The VCC pin is connected to the 5V supply of the NodeMCU, while the GND pin is connected to the ground. The TRIG pin is connected to a digital output pin (such as D1), and the ECHO pin is connected to a digital input pin (such as D2). Since the NodeMCU operates at 3.3V logic level, a voltage divider is used for the ECHO pin to ensure safe operation.

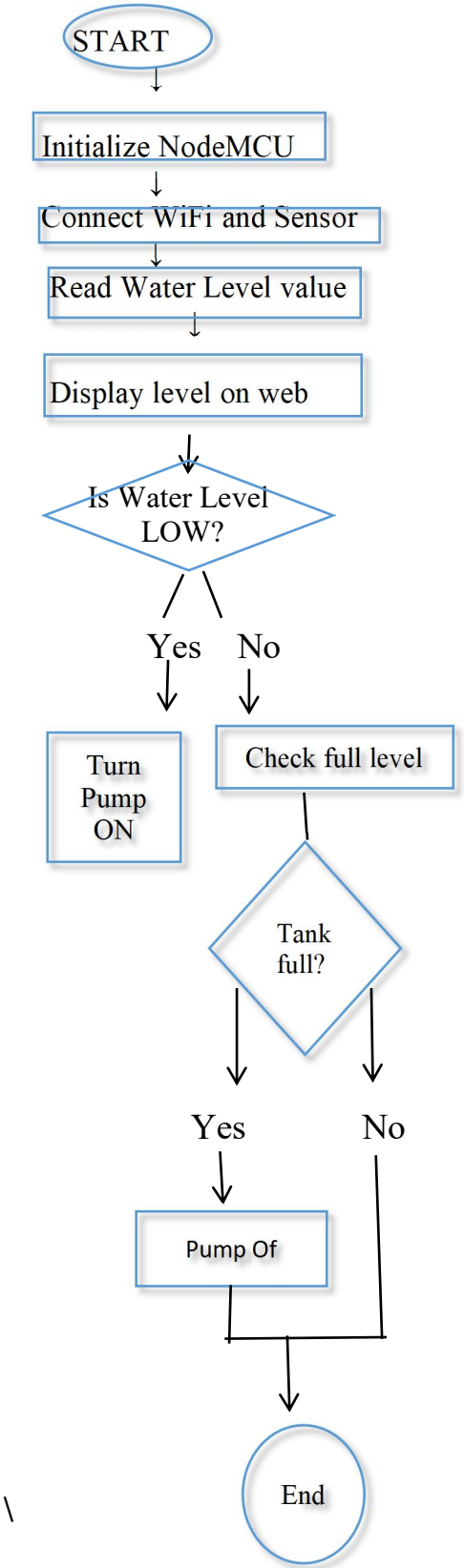
The relay module is used to control the water pump. The VCC and GND pins of the relay are connected to the power supply, and the input pin (IN) is connected to a digital pin of the NodeMCU (such as D5). The relay acts as an electrically operated switch that controls the motor.

The water pump is connected through the relay. The live wire of the power supply is connected to the COM terminal of the relay, and the NO (Normally Open) terminal is connected to the pump. The neutral wire is directly connected to the pump. When the relay is activated, the circuit is completed, and the pump turns ON.

The NodeMCU is powered using a USB cable or a regulated 5V power supply, and it also provides power to the connected components.

CHAPTER 7

FLOWCHART



CHAPTER 8

ALGORITHM

The algorithm of the proposed IoT Water Level Monitoring and Control System explains the step-by-step operation of monitoring the water level, displaying data on the web page, and controlling the water pump automatically and manually.

Step 1: Start the System

- Switch ON the power supply.
- Initialize the microcontroller system.
- Activate all connected modules and components.

Step 2: Initialize Hardware Components

The microcontroller initializes the following components:

- Water level sensor / ultrasonic sensor
- Wi-Fi module
- Relay module
- Web server
- Input and output pins
- Motor pump status

The system checks whether all modules are functioning properly.

Step 3: Connect to Wi-Fi Network

- The microcontroller connects to the configured Wi-Fi network.
- After successful connection, the IP address is generated.
- The web server becomes active for remote monitoring and control.

If the Wi-Fi connection fails:

- Retry connection continuously until successful connection is established.

Step 4: Measure Water Level

- The ultrasonic sensor sends ultrasonic waves toward the water surface.
- The reflected echo signal is received by the sensor.
- The time taken for the signal to return is measured.

Distance calculation:

Distance=Speed of Sound×Time2\text{Distance} = \frac{\text{Speed of Sound} \times \text{Time}}{2}Distance=2Speed of Sound×Time

Water level calculation:

Water Level=Tank Height−Measured Distance\text{Water Level} = \text{Tank Height} - \text{Measured Distance}Water Level=Tank Height−Measured Distance

The water level percentage is then calculated.

Step 5: Process Sensor Data

The microcontroller processes the measured data and classifies the tank condition as:

- Empty
- Low
- Medium
- Full

Example:

- 0–25% → Low
- 26–75% → Medium
- 76–100% → Full

Step 6: Upload Data to Web Page

The processed water level information is sent to the web server through Wi-Fi.

The web page displays:

- Current water level
- Water percentage
- Tank condition
- Motor status (ON/OFF)
- Manual control buttons

The web page updates continuously in real time.

Step 7: Manual Pump Control Through Web Page

The system continuously checks the user input from the web page.

Condition 1: ON Button Pressed

If the user presses the ON button:

1. Command is sent to the microcontroller.
2. Relay module is activated.
3. Water pump motor turns ON.
4. Motor status updates on the web page.

Condition 2: OFF Button Pressed

If the user presses the OFF button:

1. Command is sent to the microcontroller.
2. Relay module is deactivated.
3. Water pump motor turns OFF.
4. Web page updates motor status.

Step 8: Automatic Water Pump Control

The controller automatically operates the pump based on predefined water level limits.

Automatic ON Condition

If:

- Water level < Minimum threshold

Then:

- Relay turns ON
- Motor pump starts automatically

Automatic OFF Condition

If:

- Water level ≥ Maximum threshold

Then:

- Relay turns OFF
- Motor pump stops automatically

This process prevents:

- Tank overflow
- Water wastage
- Dry running of the motor

Step 9: Continuous Monitoring

The system continuously repeats:

- Water level measurement
- Data processing
- Web page updating
- Pump control operation

The monitoring process runs continuously in real time.

Step 10: Error Handling

If any abnormal condition occurs:

- Sensor failure
- Wi-Fi disconnection
- Power interruption

The system attempts reconnection or displays an error message on the web page.

Step 11: Stop the System

- The process stops only when the power supply is turned OFF.

Flow of Operation

Sensor Reading → Data Processing → Web Page Display → Manual/Automatic Decision → Relay Operation → Pump Control

CHAPTER 9

IMPLEMENTATION

9.1 METHODOLOGY

The methodology of the IoT Based Water Level Monitoring System explains the step-by-step approach used to design, develop, and operate the system. It focuses on how hardware and software components interact to achieve automated water level monitoring and control.

System Design Approach

The system is designed using an IoT-based embedded approach, where NodeMCU (ESP8266) acts as the main controller. The ultrasonic sensor is used to detect water level, and the relay module controls the water pump based on sensor readings. The collected data is sent to a web server for real-time monitoring.

Data Acquisition Method

The ultrasonic sensor is placed at the top of the tank.
It sends ultrasonic waves and receives the reflected signal.
The time taken for echo return is used to calculate the distance of water level.
This distance represents the water level in the tank.

Data Processing Method

NodeMCU reads sensor values continuously.
The distance value is converted into water level condition.
The system compares the value with predefined threshold levels:
Low water level → Pump ON
Full tank → Pump OFF

Control Method

A relay module is used as a switching device.
NodeMCU sends ON/OFF signals to the relay.
The relay controls the water pump accordingly.
This enables automatic water management without manual intervention.

Communication Method

NodeMCU connects to a Wi-Fi network.
A web server is created using ESP8266 library.
Real-time data is displayed on a web page using the IP address.
Users can monitor water level remotely from mobile or laptop.

CHAPTER 10

CODING

10.1 Introduction to Coding Phase

The coding phase is an important part of the IoT Based Water Level Monitoring System project. In this phase, programming is done using Arduino IDE software to control and monitor the complete system. The NodeMCU ESP8266 microcontroller is programmed to read the water level data from the ultrasonic sensor and perform automatic operations based on the detected level.

The code is developed using Embedded C language in Arduino IDE. The ultrasonic sensor continuously measures the distance between the sensor and water surface. Based on the measured distance, the NodeMCU controls the relay module to switch the water pump ON or OFF automatically. A buzzer is also activated during low water level conditions to provide an alert indication.

Additionally, WiFi functionality is implemented in the code to enable IoT features. The NodeMCU connects to the internet and creates a web server that displays the current water level status on a webpage. Users can monitor the tank level and control the pump remotely using a mobile phone or computer.

The coding phase ensures proper communication between hardware and software components, enabling efficient automation, remote monitoring, reduced water wastage, and smart water management.

PROGRAM

```
#include <ESP8266WiFi.h>
#include <ESP8266WebServer.h>

// ================== WIFI ================
const char* ssid = "YOUR_WIFI_NAME";
const char* password = "YOUR_WIFI_PASSWORD";

// ================== SERVER ==============
ESP8266WebServer server(80);

// ================== PINS ================
#define TRIG D5
#define ECHO D6
#define RELAY D1
#define LEDPIN D2

// ================ VARIABLES ============
long duration;
int distance;

bool manualMode = false;
bool pumpState = false;

String webpage;

// ======================================
// SETUP
// ======================================

void setup() {

  Serial.begin(115200);

  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);

  pinMode(RELAY, OUTPUT);
  pinMode(LEDPIN, OUTPUT);

  digitalWrite(RELAY, HIGH);
  digitalWrite(LEDPIN, LOW);

  // WiFi Connect
  WiFi.begin(ssid, password);

  Serial.print("Connecting");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
```

```
  Serial.println("");
  Serial.println("WiFi Connected");

  Serial.print("IP Address: ");
  Serial.println(WiFi.localIP());

  // Web Routes
  server.on("/", handleRoot);

  server.on("/on", pumpON);

  server.on("/off", pumpOFF);

  server.begin();

  Serial.println("Server Started");
}

// ========================================
// LOOP
// ========================================

void loop() {

  // Ultrasonic Reading
  digitalWrite(TRIG, LOW);
  delayMicroseconds(2);

  digitalWrite(TRIG, HIGH);
  delayMicroseconds(10);

  digitalWrite(TRIG, LOW);

  duration = pulseIn(ECHO, HIGH);

  distance = duration * 0.034 / 2;

  Serial.print("Distance: ");
  Serial.println(distance);

  // ======================================
  // AUTOMATIC MODE
  // ======================================

  if (!manualMode) {

    // Water LOW
    if (distance > 15) {

      digitalWrite(RELAY, LOW);

      digitalWrite(LEDPIN, HIGH);

      pumpState = true;
```

```
 }

   // Water FULL
   else {

     digitalWrite(RELAY, HIGH);

     digitalWrite(LEDPIN, LOW);

     pumpState = false;
   }
 }

 server.handleClient();

 delay(1000);
}

// ========================================
// PUMP ON
// ========================================

void pumpON() {

 manualMode = true;

 digitalWrite(RELAY, LOW);

 digitalWrite(LEDPIN, HIGH);

 pumpState = true;

 server.sendHeader("Location", "/");

 server.send(303);
}

// ========================================
// PUMP OFF
// ========================================

void pumpOFF() {

 manualMode = true;

 digitalWrite(RELAY, HIGH);

 digitalWrite(LEDPIN, LOW);

 pumpState = false;

 server.sendHeader("Location", "/");

 server.send(303);
}
```

```
// ========================================
// WEB PAGE
// ========================================

void handleRoot() {

  String waterLevel = "";
  String pumpStatus = "";

  String levelColor = "";
  String pumpColor = "";

  // Water Status
  if (distance > 15) {

    waterLevel = "LOW";

    levelColor = "#ff9800";
  }

  else {

    waterLevel = "FULL";

    levelColor = "#2196f3";
  }

  // Pump Status
  if (pumpState) {

    pumpStatus = "ON";

    pumpColor = "#ff4444";
  }

  else {

    pumpStatus = "OFF";

    pumpColor = "#00c853";
  }

  webpage = R"=====(

<!DOCTYPE html>
<html>

<head>

<title>Smart Water System</title>

<meta name="viewport" content="width=device-width, initial-scale=1">
```

```
<meta http-equiv="refresh" content="2">

<style>

body{
margin:0;
padding:0;
font-family:Arial;
background:linear-gradient(135deg,#141e30,#243b55);
color:white;
text-align:center;
}

.container{
padding-top:20px;
}

.title{
font-size:32px;
font-weight:bold;
margin-bottom:20px;
}

.card{
width:330px;
margin:auto;
padding:25px;
border-radius:20px;
background:rgba(255,255,255,0.1);
backdrop-filter:blur(10px);
box-shadow:0 0 20px rgba(0,0,0,0.4);
}

.status{
padding:15px;
margin-top:15px;
border-radius:15px;
font-size:24px;
font-weight:bold;
}

.distance{
font-size:28px;
font-weight:bold;
margin-bottom:15px;
}

button{
width:120px;
padding:15px;
margin:10px;
border:none;
border-radius:12px;
font-size:20px;
```

```
font-weight:bold;
color:white;
cursor:pointer;
}

.onbtn{
background:#00c853;
}

.offbtn{
background:#ff4444;
}

.footer{
margin-top:20px;
opacity:0.7;
font-size:14px;
}

</style>

</head>

<body>

<div class="container">

<div class="title">
SMART WATER LEVEL
</div>

<div class="card">

<div class="distance">
Distance: )=====";

  webpage += distance;

  webpage += R"=====( cm
</div>

<div class="status" style="background:)=====";

  webpage += levelColor;

  webpage += R"=====(;">
Water Level: )=====";

  webpage += waterLevel;

  webpage += R"=====(
</div>

<div class="status" style="background:)=====";
```

```
  webpage += pumpColor;

  webpage += R"=====(;">
Pump Status: )=====";

  webpage += pumpStatus;

  webpage += R"=====(
</div>

<br>

<a href="/on">
<button class="onbtn">ON</button>
</a>

<a href="/off">
<button class="offbtn">OFF</button>
</a>

<div class="footer">
NodeMCU IoT Project
</div>

</div>

</div>

</body>
</html>

)=====";

  server.send(200, "text/html", webpage);
}
```

CHAPTER 11

TESTING

11.1 Introduction

Testing is an important phase in the development of the IoT Based Water Level Monitoring System. It ensures that all hardware and software components are working correctly and the system performs as expected in real-time conditions.

11.2 Testing Objectives

To verify correct working of ultrasonic sensor.
To check Wi-Fi connectivity of NodeMCU.
To ensure proper relay operation for pump control.
To validate real-time data display on web page.
To test automatic ON/OFF control logic.

11.3 Hardware Testing

1. Sensor Testing
The ultrasonic sensor (HC-SR04) was tested by placing it at different distances from the water surface. The sensor correctly measured the distance and sent accurate readings to NodeMCU.

2. Relay Testing
The relay module was tested by sending ON/OFF signals from NodeMCU. It successfully switched the water pump ON when the water level was low and OFF when the tank was full.

1. Wi-Fi Module Testing
NodeMCU was connected to a Wi-Fi network. The IP address was generated successfully, and the system was accessible through a mobile or laptop browser.

11.4 Software Testing

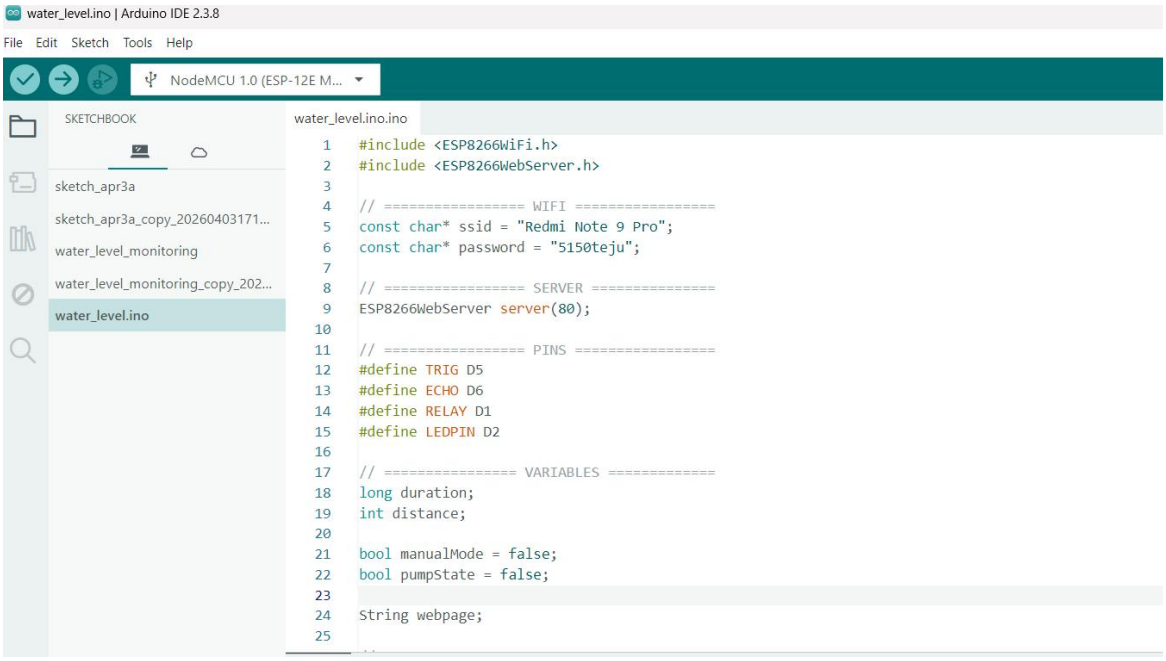
1. Code Upload Testing
The program was uploaded using Arduino IDE without errors. The system started executing immediately after reset.

2. Web Page Testing
The web page was tested using the IP address provided by NodeMCU. It displayed:
Water level distance
Pump status (ON/OFF)

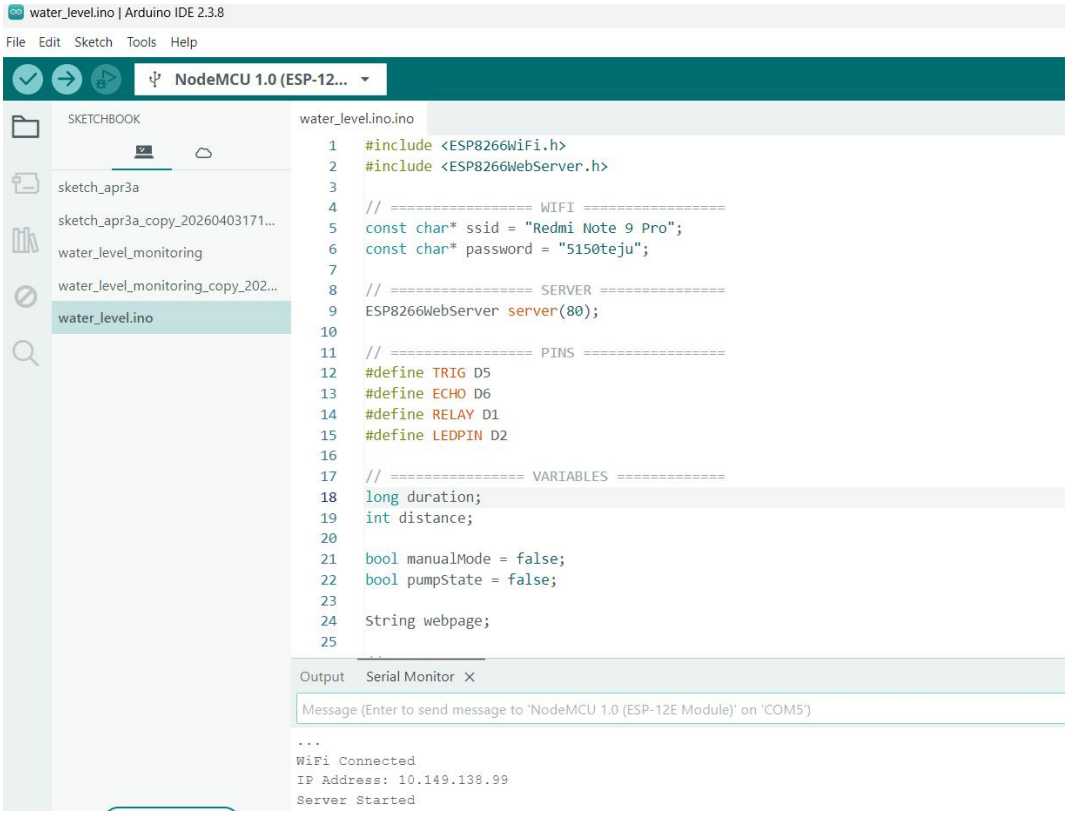
3. Logic Testing
Different water levels were simulated to test system behavior:
Low water level → Pump ON
High water level → Pump OFF

CHAPTER 12

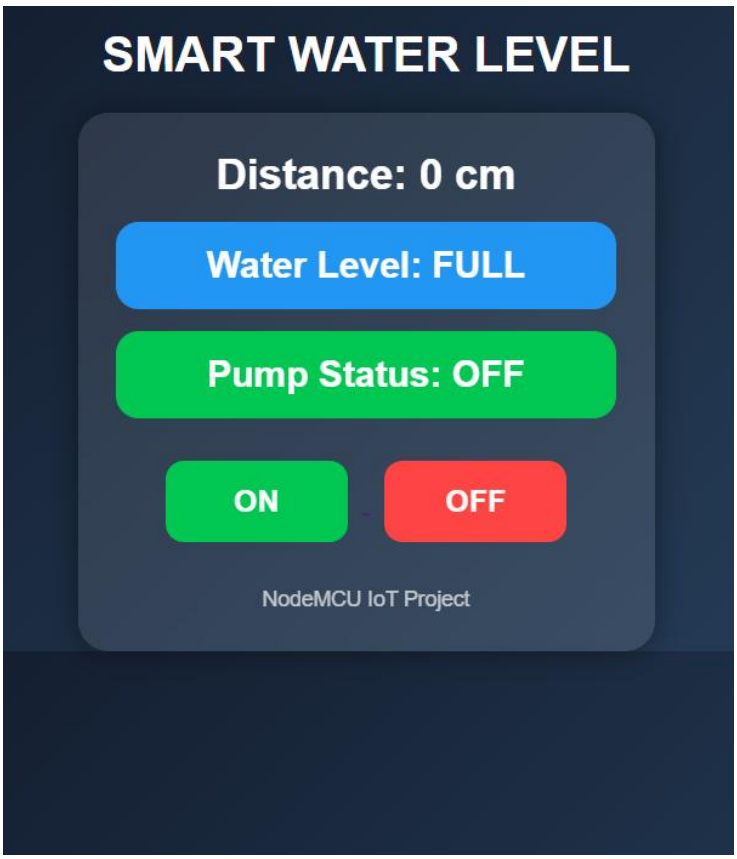
SNAPSHOT



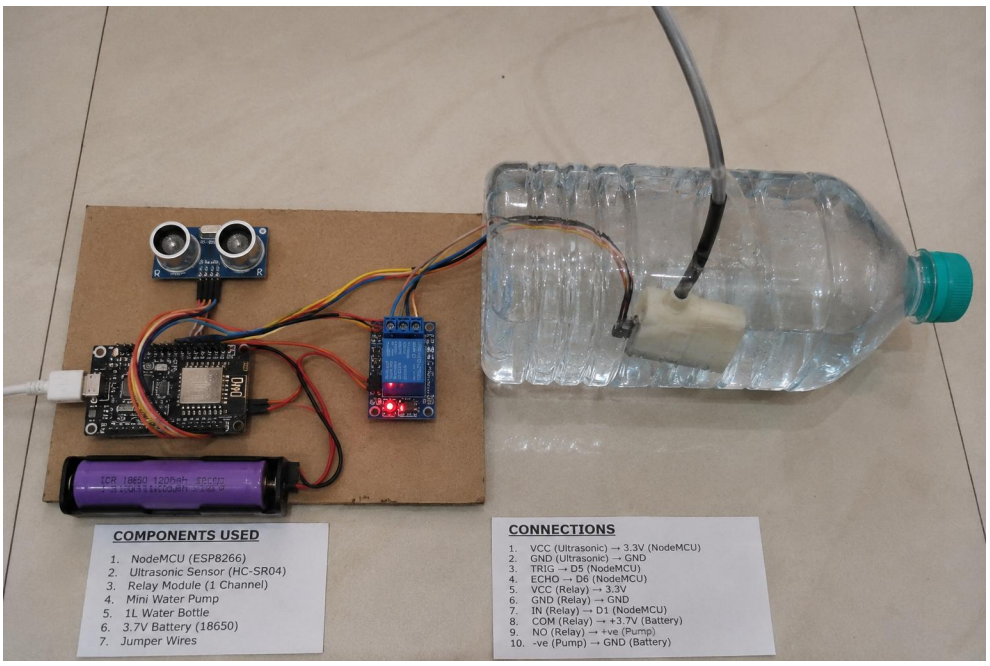
12.1 Arduino IDE



12.2 WI FI Connected



12.3 Web Page



12.4 Connection Diagram

CHAPTER 13

FUTURE ENHANCEMENT

1. Integration of mobile application support for easier monitoring.
2. Addition of GSM/SMS notification system.
3. Implementation of solar-powered operation for energy efficiency.
4. AI-based prediction of water usage and consumption.
5. Monitoring and controlling multiple water tanks simultaneously.
6. Integration with voice assistants such as Alexa or Google Assistant.
7. Addition of water quality monitoring sensors.
8. Development of advanced graphical dashboards and analytics.
9. Battery backup system for uninterrupted operation during power failures.
10. Integration with smart home automation systems.

CHAPTER 14

CONCLUSION

The IoT-Based Water Level Monitoring System successfully provides an efficient and automated solution for monitoring and controlling water levels in tanks. The system continuously measures the water level using an ultrasonic sensor and automatically controls the motor to prevent overflow and water shortage conditions.

The use of IoT technology enables real-time monitoring and remote access through a web page, making the system convenient and user-friendly. The project helps in reducing water wastage, minimizing manual effort, and improving effective water management.

Overall, the proposed system is reliable, cost-effective, and suitable for domestic, industrial, and agricultural applications. It demonstrates how IoT technology can be used for smart and efficient water resource management.

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